

CHAPTER II : HUMAN BODY AND ORGAN SYSTEM

Topic:-

Collect information about various modern treatments on heart diseases.

Introduction:-

The human heart is the most hardworking and vital muscular pump in our body. Located slightly to the left in the chest cavity between our two lungs, this fist-sized organ contracts and relaxes rhythmically around 72 times every single minute, beating over 1,00,000 times a day without taking a second of rest throughout our entire life.

The heart pumps oxygen-rich blood through arteries to nourish every single cell in our brain, muscles, and organs, and returns deoxygenated blood to the lungs. However, modern sedentary lifestyles, consumption of oily fast food, mental stress, diabetes, high blood pressure, and smoking have led to a dangerous spike in heart ailments. When coronary arteries supplying the heart muscle become clogged with cholesterol fat plaques, the patient suffers an ischemic heart attack. Fortunately, medical science has invented miraculous modern treatments that save millions of heart patients. In this project, we surveyed and collected detailed information on modern cardiac therapies.

❖ Major Types of Heart Diseases:-

- **Coronary Artery Disease (CAD):** Narrowing of heart blood vessels by fatty plaques.
- **Arrhythmia:** Irregular, abnormally fast, or dangerously slow heartbeat.
- **Heart Failure:** Weakened heart muscle unable to pump sufficient blood volume.

Aim & Medical Survey Preparation:-

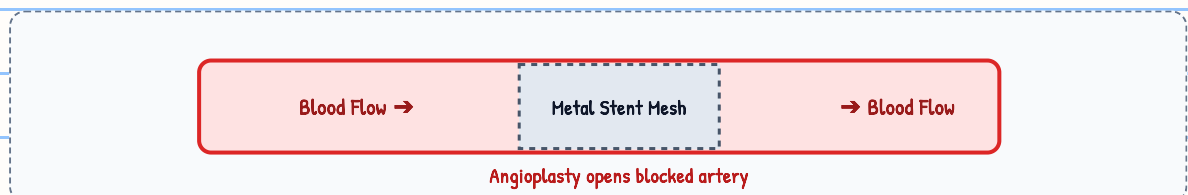
Aim: To collect, organize, and study authentic medical information regarding modern therapeutic and surgical treatments for cardiovascular heart illnesses.

❖ *Objectives of the Study:-*

- To understand how minimally invasive Angioplasty and Stenting open blocked arteries.
- To examine Coronary Artery Bypass Graft (CABG) open-heart surgery.
- To study electronic Pacemakers and Heart Transplant procedures.

❖ *Sources of Information Researched:-*

- Class 8 General Science textbook (Chapter 11: Human Body and Organ System).
- Interview with a practicing cardiologist at our district multispecialty hospital.
- National Health Portal of India (nhp.gov.in) cardiovascular education brochures.



Overview of Modern Heart Treatments:-

Our medical research documented four revolutionary modern heart treatments:

- **1. Angioplasty & Stenting (Minimally Invasive):** Instead of opening the chest, a cardiologist inserts a thin flexible catheter tube through a blood vessel in the patient's wrist or groin. A tiny balloon at the tip is inflated to crush the fat plaque against the artery wall. A spring-like metal mesh tube called a '*Stent*' is permanently expanded in place to keep the artery wide open.
- **2. Coronary Artery Bypass Graft (CABG Surgery):** When multiple coronary arteries are completely blocked, surgeons take a healthy clean blood vessel from the patient's leg (saphenous vein) or chest and graft it around the blockage, creating a brand new detour road for blood!
- **3. Electronic Pacemakers:** A matchbox-sized battery-operated computer implanted beneath the skin under the collarbone. Thin wires connect to heart chambers to deliver tiny electrical impulses that keep an irregularly beating heart steady at 72 bpm.
- **4. Heart Transplantation:** In cases of terminal heart failure, surgeons replace the diseased heart with a healthy donor heart from a brain-dead organ donor.

Treatment Comparison Table:-

The table below compares the procedures, indications, and recovery parameters of cardiac treatments:

<i>Treatment Modality</i>	<i>Clinical Condition Treated</i>	<i>Surgical Approach</i>	<i>Hospital Stay & Recovery</i>
Angioplasty & Stent	1 or 2 coronary artery blockages	Keyhole catheter via wrist artery	1 to 2 days; resumes work in a week
Bypass Surgery (CABG)	Severe triple-vessel blockages	Open-heart bypass vascular graft	7 to 10 days; 6 to 8 weeks recovery
Artificial Pacemaker	Bradycardia (abnormally slow heart rate)	Minor chest implant under local anesthesia	1 day stay; battery lasts 10 to 15 years
Heart Valve Replacement	Leaking or stiff heart valves	Mechanical or biological tissue valve	5 to 7 days; lifelong medication
Heart Transplant	End-stage congestive heart failure	Total organ replacement surgery	3 to 4 weeks; requires immunosuppressants

Scientific Analysis & Diagnostic Tools:-

Modern cardiology relies on advanced non-invasive diagnostic tools before choosing surgery:

- **Electrocardiogram (ECG):** Electrodes placed on the chest record the electrical conduction waves (P-Q-R-S-T) of heart contractions. An abnormal spike instantly reveals a heart attack in progress.
- **2D Echocardiography (Echo):** Uses harmless ultrasound waves to display live moving video of heart chambers, pumping strength (Ejection Fraction), and leaking valves.
- **Coronary CT Angiography:** Advanced 128-slice CT scanners capture 3D X-ray maps of heart blood vessels in 10 seconds without any needle puncture, detecting blockages early.

❖ The Golden Hour in Heart Attacks:-

The first 60 minutes after severe chest pain starts is termed the 'Golden Hour'.

Administering clot-dissolving medications (Thrombolysis) or performing emergency angioplasty within this hour prevents permanent heart muscle death!

Prevention & Healthy Lifestyle Habits:-

Doctors emphasize that 80% of premature heart attacks can be prevented through simple habits:

- **1. Balanced Heart-Healthy Diet:** Cut down deep-fried foods, trans-fats, processed sugars, and excess salt. Eat fiber-rich oats, leafy greens, walnuts, and fresh seasonal fruits.
- **2. 30 Minutes of Daily Aerobic Exercise:** Brisk walking, cycling, swimming, and skipping strengthen the heart muscle and keep coronary blood vessels flexible.
- **3. Stress Reduction & Pranayama:** Deep breathing exercises and yoga lower adrenaline stress hormones that spike blood pressure.
- **4. Zero Tobacco Policy:** Smoking and chewing tobacco constrict blood vessels instantly and increase heart attack risk by 400%!

❖ Free Emergency Ambulance Helpline:-

Dial **108** in Maharashtra for immediate free emergency cardiac ambulance support with on-board oxygen and trained paramedics.

What We Learned & Conclusion:-

- The heart is a vital muscular pump working continuously to supply oxygenated blood to the body.
- Fatty cholesterol deposits cause coronary blockages, resulting in life-threatening heart attacks.
- Modern medicine has miraculous treatments like angioplasty, stents, bypass surgery, and pacemakers.
- Eating a balanced diet, exercising daily, avoiding tobacco, and controlling stress protect our heart naturally.

Conclusion:-

Modern medical treatments for heart diseases represent one of the greatest triumphs of clinical biomedical engineering. Procedures like stent angioplasty and electronic pacemakers give heart patients a healthy second lease of life within days. However, medical technology cannot substitute for healthy living.

Maintaining daily exercise and nutritious food habits remains the true, lifelong key to a strong and joyful heart.

References:-

- Maharashtra State Board Class 8 General Science Textbook (Chapter 11: Human Body and Organ System).
- Cardiological Society of India (CSI) Patient Educational Series on Heart Health.
- National Health Portal of India (nhp.gov.in) – Cardiovascular Health Guidelines.